Game systems

Virtual pet



Welcome to Game Systems!

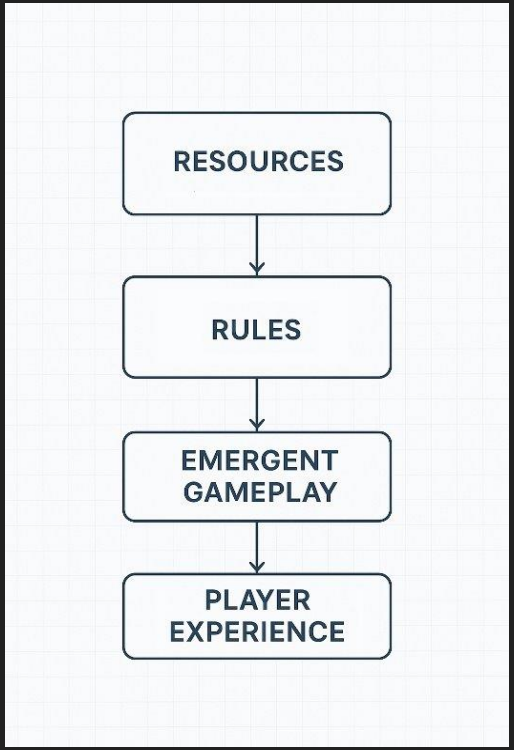
In this class, you will learn to:

- Work with Systems: Learn to think in terms of resources, rules, and states. We'll explore how simple parts can create complex, interesting behaviors.
- Balance Economies: Use tools like Machinations to simulate and balance your game's economy, ensuring your loops keep players engaged.
- Create Emergent Gameplay: Discover how to use simple rules to create behaviors and situations you didn't even plan for. This is where the magic of game design happens.
- Technical Prototyping: Get your hands dirty. You will configure components, work with state machines, create interactive diagrams and work with a framework of Unity Events to build and test your prototypes, turning theory into playable experiences.



From resources to player experience

At its core, a game is a system. Resources and rules set the stage, and players explore what’s possible within them. The real depth comes from how these simple elements combine, leading to unexpected outcomes.

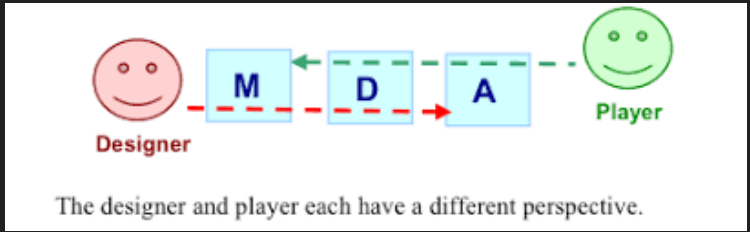


MDA Framework

This model is a direct parallel to the MDA framework. We set up and configure mechanics to shape the player's aesthetic (the experience).

The game's dynamics are what happen in between, the emergent behaviors that come from a player interacting with the game's mechanics.

This way of looking at it helps you move from a feeling you want to create to the actual systems you need to build.



The designer and player each have a different perspective.

Course overview

This course has three assignments:

- Virtual pet
- Progression game
- Sandbox

Virtual Pets



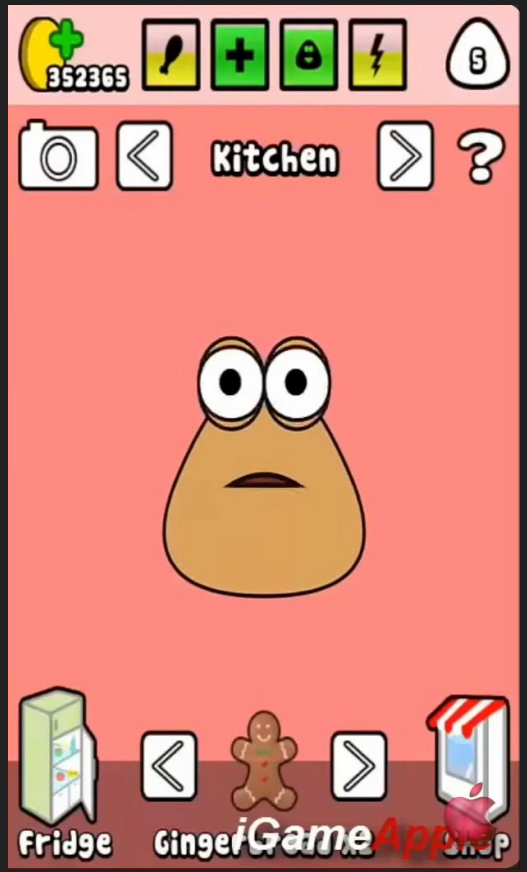
Pou

Pou (/puː/ or /poʊ/) is a 2012 virtual pet mobile game for BlackBerry 10, iOS and Android developed by Lebanese designer Paul Salameh and published under the name Zakeh Limited to the Google Play Store and the Apple App Store. The game plays similarly to Tamagotchi, a fad game that requires caring for a simulated creature.[1][2] It was one of the most downloaded games of 2014.[3]

Our first assignment is inspired by virtual pet games such as Pou.

° Who has played Pou?

° How would you describe Pou's personality?



What is a Virtual Pet?



A virtual pet is a digital entity that mimics life-like behavior and requires care from the player.

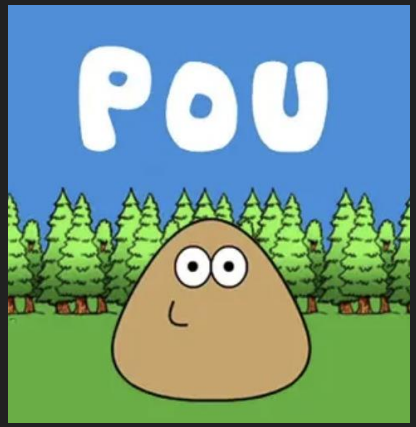
These pets respond to player actions and simulate emotional needs, often forming a bond with the player.

- A simulated creature with needs and personality.
- Examples: Tamagotchi, Pou, Nintendogs, Neopets.
- Designed for mobile or handheld platforms.
- Responds to feeding, playing, and attention.

Examples


TAMAGOTCHI


neopets


Pou


VOLUME 1
DIGIMON
THE OFFICIAL
01




Date
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Trouble

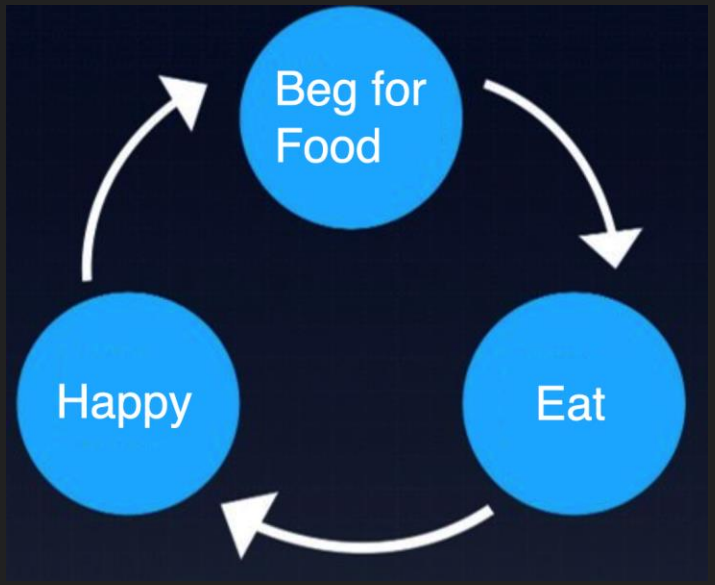



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The Core Gameplay Loop

The core gameplay loop defines the repetitive structure of interaction. It creates rhythm and expectation and helps players form habits around caring for their pet.

- Feed pet → Pet reacts → Mood changes → Player responds → Repeat.
- The loop must be short, intuitive, and emotionally rewarding.
- Feedback should reinforce player actions.



The Core Questions: Unlocking Player Engagement



It's not just about what the game *does*, **but *why* players care**. Understanding these reasons helps you make a game that truly resonates.

- Why do people play virtual pet games? What's the appeal?
- What makes players feel connected or entertained?
- How can we amplify those positive feelings and effects?
- How do we give a 'machine' a personality players will love?

The Game Designer's Toolkit



Game design stands on strong foundations from different fields.

You'll draw on insights from philosophy and psychology to understand human nature, use science and tech to build the mechanics, and apply art and design to craft the overall experience. Your task is to combine these to create something truly special.

- Philosophy & Psychology: Understanding *why* players connect and what makes them tick.
- Science & Tech: Building the game world and its behaviors through clever coding and systems.
- Art & Experience: Crafting the look, feel, and emotional journey of the player.

Philosophy: What is Life?

A virtual pet isn't truly alive, but as a designer, you have to make it *feel* that way. This is where philosophy comes in. You're exploring big questions about the nature of a relationship between a human and a machine.

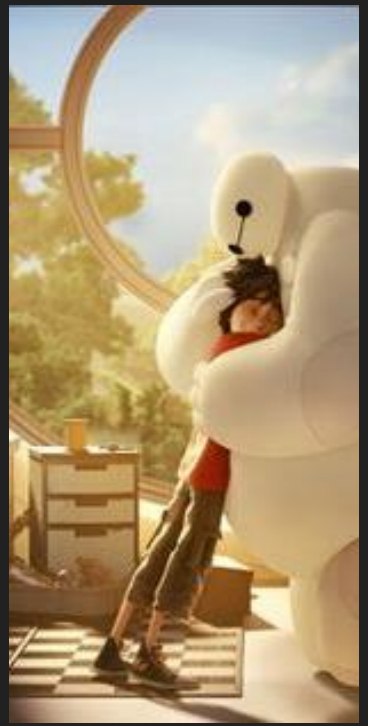
* What defines a 'living' being in a virtual world?
* What is the difference between a character and a pet?
* How do we get the player to believe this digital creature has real needs and feelings?



Building a Real Bond

The magic of a virtual pet lies in the player's connection to it. Psychology helps you understand the emotional triggers that make this possible. You are essentially designing an emotional bond.

- What makes us feel empathy or a sense of responsibility for something that isn't real?
- How can you use rewards, routines, and needs to make the player feel the pet is needed and cared for?
- Why do we get upset when our digital pet "dies" or is unhappy?



The Illusion of Life

The code is the skeleton of your pet, but it's the design that gives it a soul. Technology is how you translate all your philosophical and psychological ideas into a functioning game.

- How can you use AI or simple algorithms to make the pet's behavior seem unique and unpredictable?
- What behaviors or reactions will make the player feel the pet is "thinking" and not just following a script?
- What technical choices can create the feeling of a genuine personality?

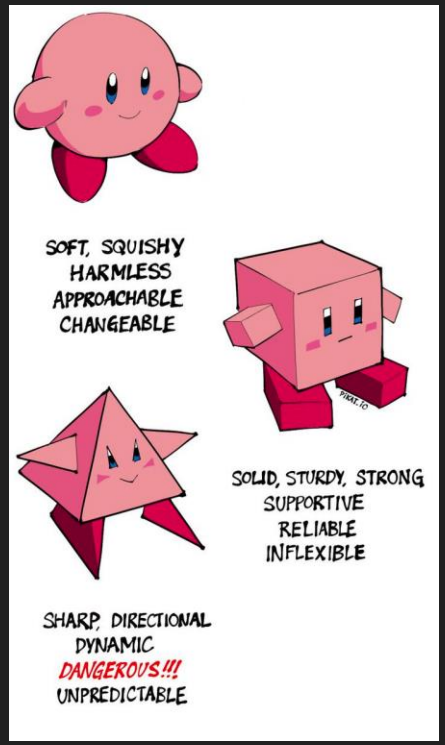


Creating your pet - Shape language

The art of using simple forms to communicate a feeling or personality.
It's a visual shortcut that the player understands instantly. There are three basic shapes:

- Squares and Rectangles: These represent strength, stability, and reliability. They often feel safe but can also come across as stiff or unyielding.
- Circles and Ovals: These communicate friendliness, innocence, and softness. Round shapes often feel approachable and warm.
- Triangles and Sharp Shapes: These suggest energy, aggression, or danger. A triangle can represent a dynamic character, but also a threatening one.

The shape of your pet, from its head to its body, already communicates something about its personality before it has even made a single move. Think about what you want your pet to communicate first.



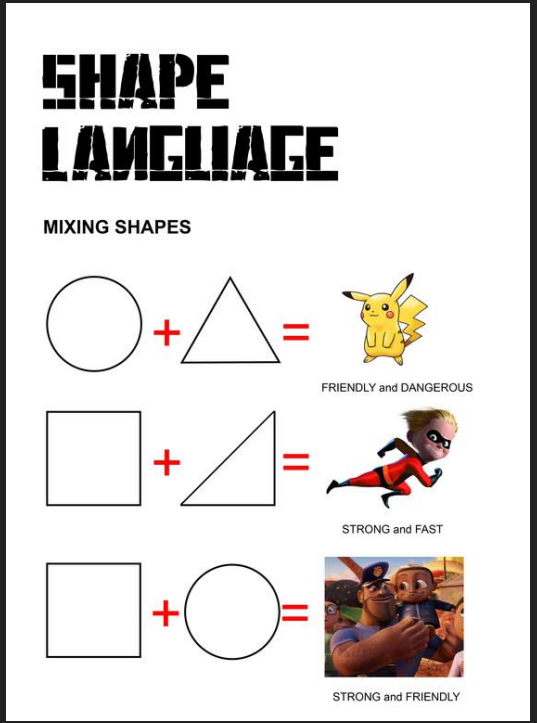
Combining shapes

A character is rarely made from just one shape. By combining shapes, you can create a more complex and interesting personality. You mix the meaning of the basic shapes to design a layered character.

How does this work?

- Dominant Shape: Use a dominant shape to show your pet's main characteristic. A playful, innocent pet, for example, might have a round body.
- Secondary Shape: Add a secondary shape to give a subtle hint of a nuance. If that same pet has sharp, triangular ears, it could suggest a mischievous or cunning side.
- Details: Even small details matter. A pet that is mostly round but has sharp teeth can hint at a hidden aggression.

By intentionally combining shapes, you can visually communicate the personality you designed earlier. You're not just creating a design; you're telling a story with shapes.



Color language basics

Just like shapes, colors are a powerful tool to instantly communicate personality. The psychology of colors has been used for centuries to evoke specific emotions. By choosing the right color for your pet, you can give the player a clue about its personality before it even moves.

Here are the basic meanings of some primary colors:

- Red: Suggests energy, passion, and excitement. A red pet might be bold and playful, but can also seem aggressive.
- Yellow: Represents cheerfulness, optimism, and warmth. A yellow pet might be happy and friendly, but can also come across as timid.
- Blue: Evokes a sense of calm, intelligence, and loyalty. A blue pet might be wise and peaceful, but could also be seen as sad or cold.

Think about the main emotion you want your pet to convey and start there.



Combining colors

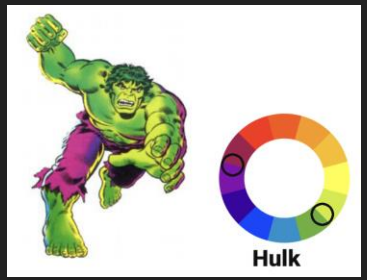
Characters are rarely just one color. By combining different colors, you can create a more complex and nuanced personality. Just like with shapes, you'll use a main color to show the pet's core personality and an accent color for a hint of something more.

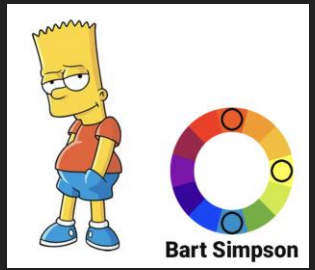
How to combine them:

- **Main Color:** Choose a primary color that represents your pet's dominant personality trait. A pet that is mostly peaceful and wise might be blue.
- **Accent Color:** Add a small amount of a contrasting color to add depth or a unique quirk. For example, giving that blue pet a small yellow detail can make it seem more optimistic and less sad. Or, a red detail could hint at a hidden, playful side.

Using color combinations is a great way to visually reinforce the personality you designed earlier. Your final design, with its blend of shapes and colors, is your pet's visual resume. It tells the player everything they need to know at a glance.









Baby Schema: The Science of Cute

When you design your virtual pet, you want players to feel a natural urge to care for it. This is not just a feeling; it's a deep-rooted biological instinct called the Baby Schema.

The baby schema is a set of features that we are biologically programmed to find cute and endearing. It triggers a parental care instinct in mammals (including humans!) and makes us want to protect and nurture the creature.

The key features are:

- A large, round head relative to the body.
- Large eyes.
- A small nose and mouth.
- A plump, rounded body and soft limbs.

By incorporating these features into your pet's design, you are tapping into a powerful drive to make your pet instantly lovable.



Baby Schema in Action

You see the baby schema everywhere in character design, from animated movies to video games. Animators at Disney, Pixar, and Studio Ghibli use these features to make their characters irresistible. Think of the big eyes on a character like Totoro or a baby Groot.

It's also a fundamental part of the kawaii aesthetic from Japan, which focuses on characters that are adorable, small, and round. Many popular franchises, like Pokémon and games like Cut the Rope, use these design principles to make their characters appealing and successful.











Making pets cute

For your virtual pet, this means you can intentionally choose shapes and proportions to create a pet that feels innocent and helpless. By doing so, you are not just designing a character—you are creating an emotional connection that encourages the player to take care of it.



Exercise Character concept art: the first pass

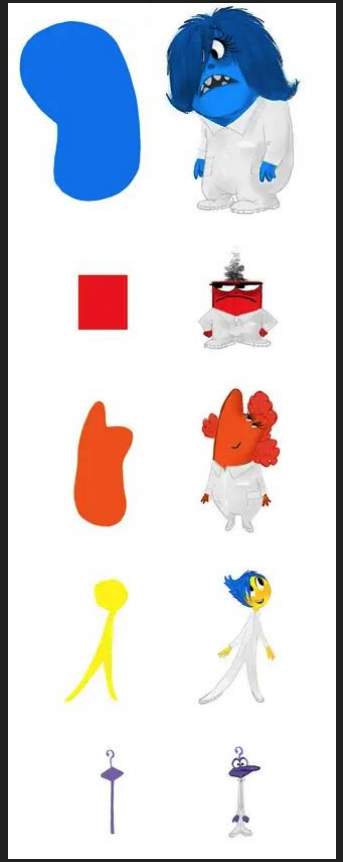
You have now defined your pet’s personality. The next step is to translate these ideas into a visual concept that you can share with an artist. Remember, this isn't about creating a perfect drawing; it’s about creating a clear blueprint.

Your task is to create a rough outline of your pet using basic shapes and colors. This sketch serves two purposes:

A Creative Blueprint: It forces you to think about how your pet's personality translates visually. Does its calm nature make it round? Does its quirky habit make its ears pointy?

A Communication Tool: This sketch is what you would hand over to an artist. It clearly shows the shapes and colors you want, making sure everyone is on the same page from the start.

Use a simple drawing tool or just a piece of paper. Focus on the core message of your design. What story do the shapes and colors tell? This is the first step in bringing your pet to life.



Personality: Bringing Your Pet to Life

So far, you've used activities and parameters to build your pet's behavior with logic. But to make your pet truly unique, you need to go beyond the rules. This is where personality comes in.

Your pet's personality is the overall way it reacts to the world. It's not a single condition you can code; it's a general feeling the player gets. Is your pet naturally friendly? Or is it shy and takes time to warm up to people?

A well-designed personality makes your pet's actions predictable in a believable way. The player learns to know their pet and can build an emotional connection because they understand why the pet does what it does.

A strong personality is the bridge between your technical FSM and the player's emotional experience.

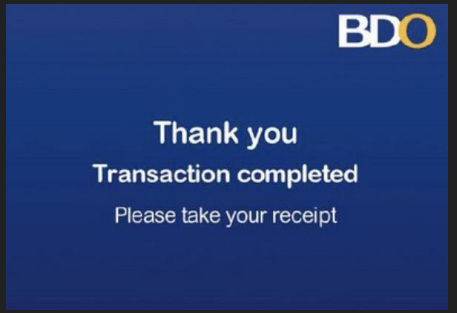


The Illusion of Personality

We often see faces or personality where there are none. This is due to two psychological phenomena:

- Pareidolia: The tendency to perceive a familiar pattern (like a face) in random or ambiguous stimuli. Our brains are hardwired to look for faces and human-like shapes, even in inanimate objects.
- Anthropomorphism: The attribution of human characteristics, emotions, and intentions to non-human entities. When a simple machine says "thank you", we instinctively project human-like gratitude onto it.

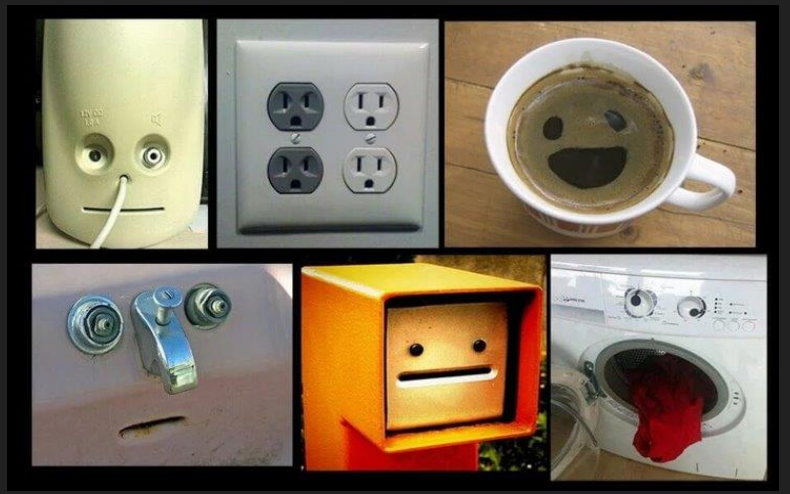
These principles show that you don't need a complex system to create the illusion of personality. You only need to provide the right cues to trigger the human brain to fill in the rest.



Pareidolia: The Illusion of a Face

You don't need to design a perfect human face to convey emotion. Even the most abstract designs, such as two dots for eyes and a curved line for a mouth, are enough to activate the human tendency for pareidolia.

Practical Example: Make your pet's eyes and mouth clearly visible and expressive. Use motion, like blinking or a subtle smile, to strengthen the illusion of life.



Anthropomorphism: Giving your pet Human Traits



Antromorphism is about projecting human communication onto an animal or being. A pet wagging its tail is animal behavior. A pet wagging its tail happily after you pet it is an anthropomorphic communication. Give your pet a 'human' body language.

- Reward and Gratitude: Make your pet react with a happy animation after the player does something. The player will unconsciously feel rewarded and appreciated. Example: The pet wags its tail, jumps up once, and then runs to you as if to 'reward' you for playing.
- Empathy and Vulnerability: Show your pet in a vulnerable state, like a hunched posture or a shivering animation when it's sad. This taps into the player's natural empathy and strengthens the bond. Example: The pet cries with its paws over its eyes, an exaggerated sad expression, or crouches down.
- Habit and Recognition: Give the pet a specific, unique habit that has no clear relation to the gameplay mechanics. For example, a pet that always spins around before going to sleep, or that shakes its head when confused. These small, seemingly useless details are the most human and create a unique character.

The goal is to attribute human emotions and intentions to your pet's movements.

Personality: A Pet That Connects

The most attractive virtual pets have personalities that trigger our natural desire to care for them. They are designed to be relatable and rewarding, making the player want to keep coming back.



- The Vulnerable Pet: Starts shy or needy. This makes the player feel important and responsible, and the pet's journey to happiness becomes a powerful reward. This idea is based on the work of Konrad Lorenz on the baby schema, which says we are biologically wired to care for creatures with cute features.
- The Expressive Pet: This pet has clear and easy-to-read emotions. Its body language, sounds, and animations are obvious. This helps the player feel understood and gives them clear feedback that their actions are working. Think of a dog's wagging tail or a cat's purr.
- The Rewarding Pet: This pet's personality is all about positive reinforcement. When the player takes good care of it, the pet responds with clear, joyful actions. This is rooted in the behavioral psychology of B.F. Skinner, where clear rewards encourage and reinforce a desired behavior.
- The Quirky Pet: A pet with a unique, sometimes silly, personality feels more like an individual. Its unpredictable quirks add a layer of surprise and make the player feel like they have a truly special friend.

The most successful virtual pets often combine these traits. Your job as a designer is to decide which traits you want to use to bring your pet to life.

Exercise: Define Your Pet's Personality

Now it's time to translate your pet's behavior and moods into a concrete personality. Use this questionnaire to think about your pet's character traits and write them down. This will be the blueprint for all of your pet's reactions in the game.

Your task is to describe your pet's personality by answering these questions:

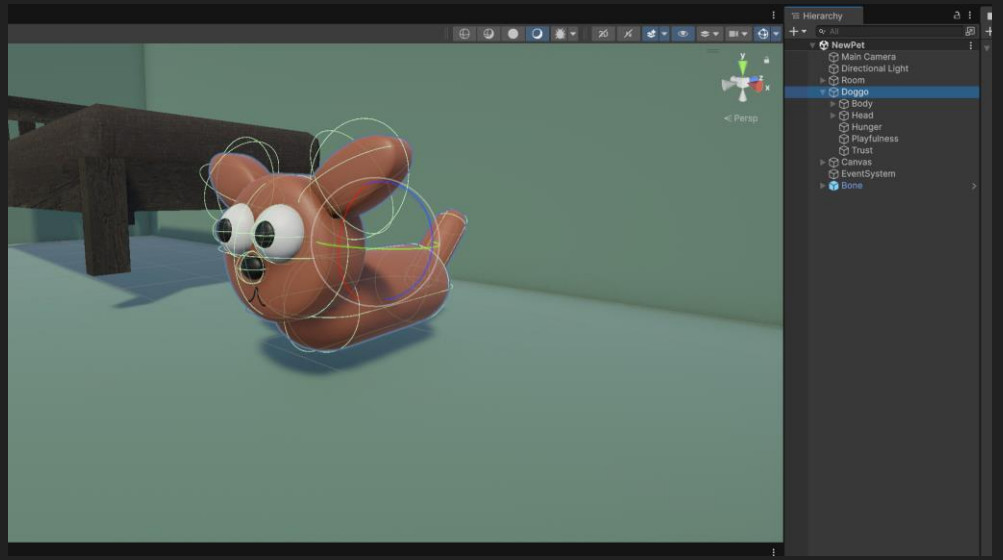
- General Disposition: What is your pet's default setting? Is it naturally energetic and playful, or is it calm and shy?
- Social Interaction: How does your pet react to the player? Is it immediately affectionate, or does the player have to earn its trust first?
- Adversity: How does your pet handle negative parameters, like high boredom or low energy? Does it get irritable and angry, or does it become sad and withdrawn?
- Success: How does your pet react when the player takes good care of it? Does it become jubilant and loud, or content and quiet?
- Unique Quirk: What is one unique behavior or unexpected trait that makes your pet stand out from all the others?

Write down a few sentences for each of these questions. This description is the foundation of your pet's personality and will help guide its behavior throughout the game.



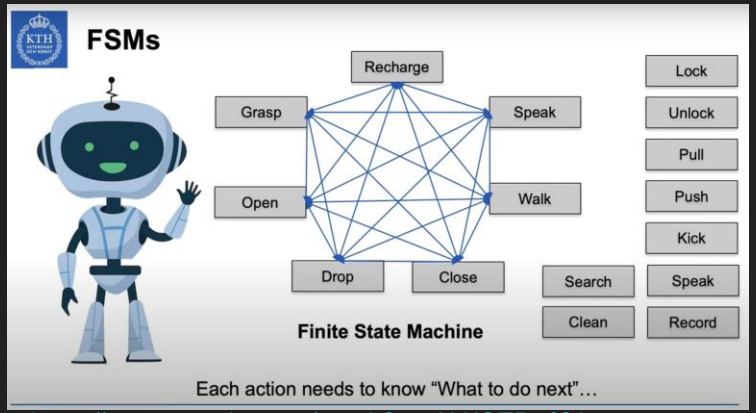
Exercise: Create your pet in Unity (30 mins)

- Follow the first half of the video: "Setting up animations" on Brightspace
- Begin creating your pet in Unity
- Use primitives as the base for the pet
  You can use primitives from Blender/Maya/Probuilder
- Think about what you are going to animate later and make use of empty parents (do not forget to think about pivot points)



Virtual Pets: shaping a personality in a technical system

- Understand how systems can simulate personality - by combining variables, states, and feedback loops
- As designers your focus is on how **non-verbal communication** (posture, timing, and behavior) can express emotion and character - create a virtual pet that feels alive and emotionally responsive.
- This design challenge is rooted in interaction, empathy, and system thinking.



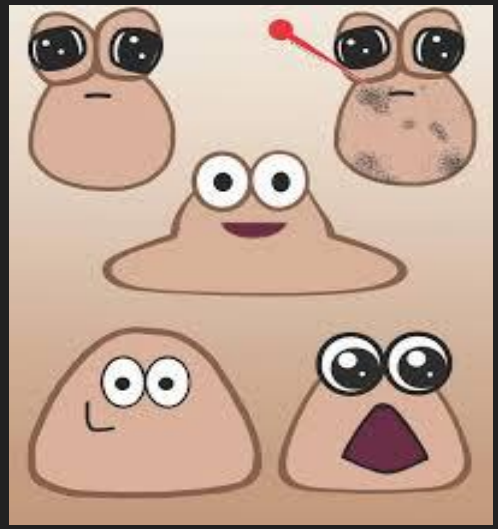
https://www.youtube.com/watch?v=gXrKGTPwfO8

States Explained

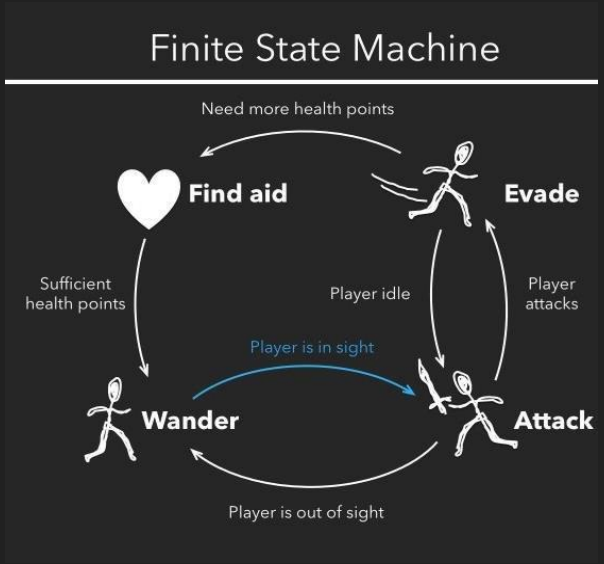
A state is a mode or condition the pet is in. States define how the pet behaves and responds to the player, and they are essential for organizing game logic.

Think of states as activities your pet can do: Eating, Sleeping, Playing

- Each state has a specific animation
- States are mutually exclusive (only one is active at a time). Transitions between states are triggered by conditions.



Design pattern: What is a Finite State Machine?



An FSM, or Finite State Machine, is a diagram-based structure for defining a character's behavior. It represents behavior as a series of discrete states or "modes." The pet can only be in one state at a time, and the FSM defines specific transitions or rules for moving from one state to another.

- Used to model pet behavior in a structured way.
- States are nodes; transitions are arrows with conditions.
- Helps to visualize and debug complex systems.

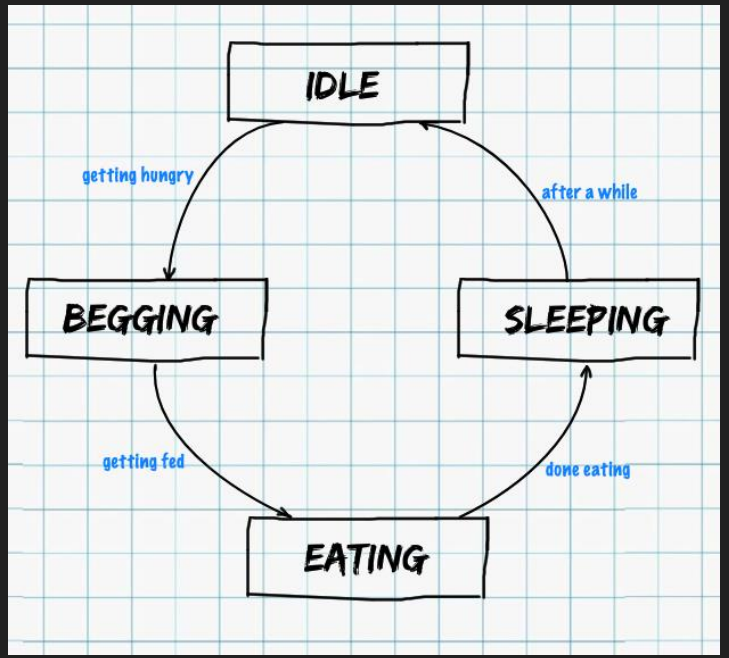
Reference: Unity Documentation on Animator Controllers.

State diagram exercise (10 mins)

This exercise will help you design the Finite State Machine (FSM) that will control your pet's actions. Think of it as drawing a simple map of what your pet can do.

- Map the Actions: Draw circles for each action your pet can take. Think about states like Idle, Eating, Sleeping, or Playing.
- Connect the Transitions: Draw arrows from one action to another to show how the pet can switch. For example, from Idle to Eating.
- Label the Conditions: Write a simple condition next to each arrow. This is the "rule" for the transition. Don't worry about numbers yet—just describe it in plain language. For example:
  -From Idle to Eating: "If it's hungry."
  -From Eating to Sleeping: "If it's full."

This exercise is about organizing your ideas. Use paper, a whiteboard, or any drawing tool—just focus on making the logic clear.



Values in Virtual Pets

The internal state of the pet is represented by values. These values track needs and drive behavior, forming the basis for emotional feedback and gameplay logic.

- Common properties: Hunger, Happiness, Energy, Boredom.
- These values typically range from 0 to 100 for intuitive control.
- Values change over time or through player actions.





What are Parameters?

When you are designing your virtual pet, you need to decide what data you are going to track. These are your parameters.

Parameters are variables that hold the current state of your pet. They are like the pet's health metrics or mood stats. For example, your pet might have parameters for:

- hunger: A number that goes up over time.
- boredom: A number that goes up if the pet is idle.
- energy: A number that goes down when the pet plays.

```
class Pet:
    def __init__(self, name):
        self.name = name
        self.hunger = 50
        self.energy = 50
        self.happiness = 50
```

https://python.plainenglish.io/how-to-build-a-virtual-pet-in-python-like-a-tamagotchi-366b5343480a

What are Conditions?

What are Conditions?

In your state diagram, the conditions are the logical rules that tell your pet when to switch from one mood to another. They are the "if this, then that" statements that bring your design to life.

Conditions are based on two things:

- Parameter values: hunger, boredom, energy, any value that changes over time. Your conditions will use these values to make decisions. For example, "If the hunger variable is greater than 80, transition to the Hungry state."
- Player actions: The player's actions, such as petting the animal or giving it food, also trigger conditions.

```
def feed(self):
    self.hunger = max(self.hunger - 20, 0)
    print(f"{self.name} has been fed!")

def play(self):
    if self.energy >= 20:
        self.happiness = min(self.happiness + 20, 100)
        self.energy -= 20
        print(f"{self.name} played and had fun!")
    else:
        print(f"{self.name} is too tired to play.")

def sleep(self):
    self.energy = min(self.energy + 30, 100)
    self.hunger = min(self.hunger + 10, 100)
    print(f"{self.name} took a nap.")
```

State diagram exercise part 2 (10 mins)

Now that you've thought about your pet's states, let's make your diagram more technical. We're going to introduce parameters: the variables that track your pet's needs.

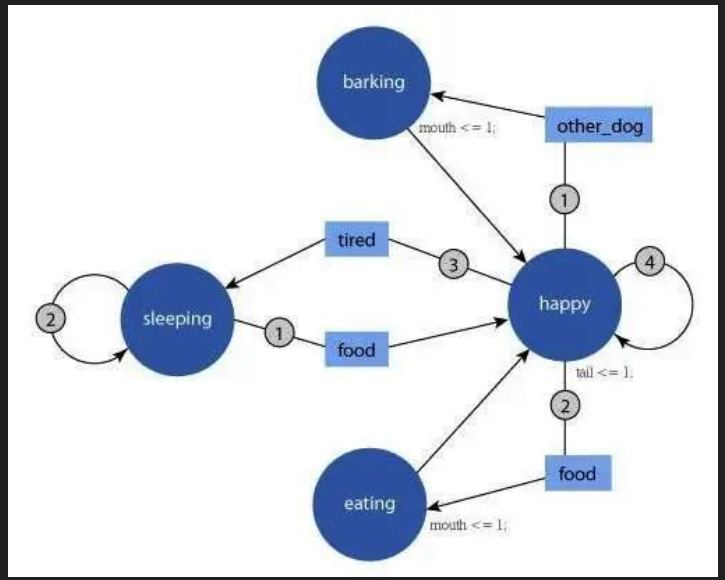
For this exercise, focus on simple parameters, such as:

- hunger: A number that goes up when your pet isn't fed.
- boredom: A number that goes up when your pet isn't played with.
- energy: A number that goes down when your pet is active.

Your task is to revisit your state diagram and label each transition with a specific parameter condition. Instead of "If it hasn't been fed in a while," your condition should now be something like:

- hunger > 80 ?→ Go to the "Hungry" state.

Think about what specific numbers or values would trigger each of your transitions. Don't worry about the exact numbers just yet—the goal is to get used to thinking in terms of data-driven conditions. This is how you'll make your pet's behavior predictable and controllable in the game.



Practice

- Follow the video "Setting up the state machine"

Moods: The Heart of the Pet

So far, you've used parameters like hunger and energy to make simple transitions. But a pet's personality is more than just a set of numbers. This is where moods come in.

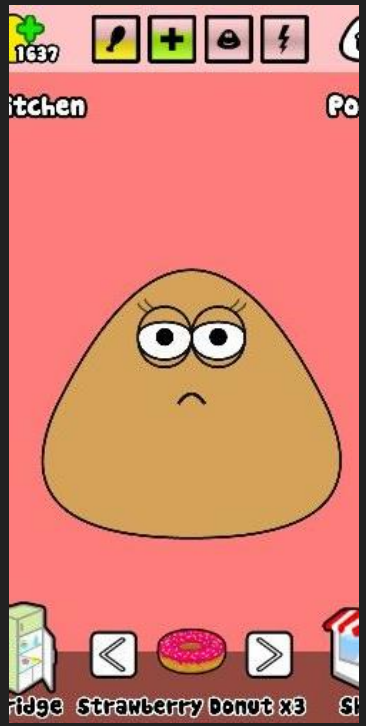
A mood is a more complex emotional state that is caused by a combination of parameters. For example, your pet's Depressed mood might be the result of a high hunger AND a low happiness for a long period of time.

Moods are what make your pet feel alive and relatable. They create a stronger connection with the player because the player can feel what the pet is feeling and understand the why behind its behavior.

Think of it this way:

- Parameters are the basic, trackable needs of your pet. They are the simple facts, like 'the pet hasn't eaten' or 'the pet is low on energy.'
- Moods are the emotional story that you build on top of those facts. They're the 'why' behind your pet's behavior. The player doesn't just see a high hunger value; they see a pet that is Depressed because it has been hungry for too long.

This is a key part of great game design—using simple facts to create a compelling emotional experience that the player can connect with.



Designing moods (15 mins)

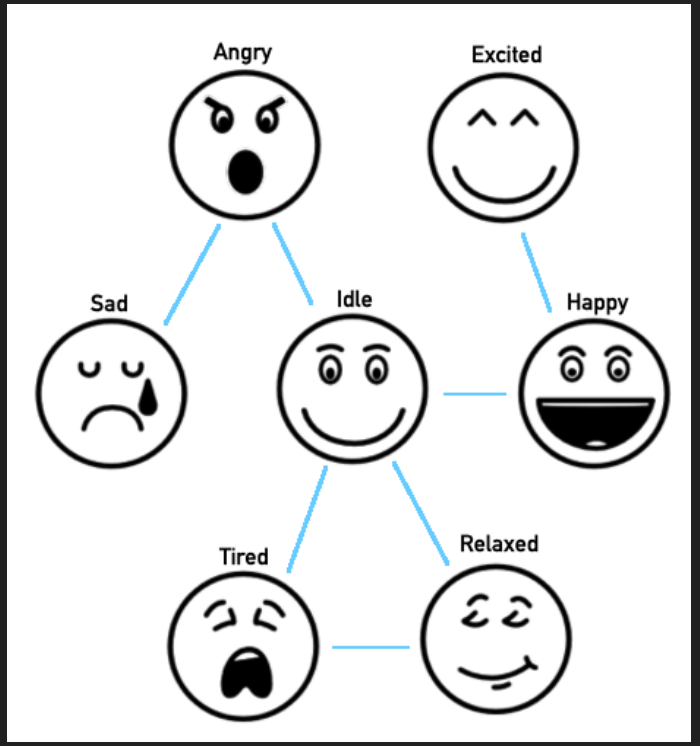
Now it’s time to take your pet's design to the next level. You are going to create a new state diagram, but this time, the states will be the pet's overarching moods, which are created by the parameters you've already defined.

Your task is to draw a new state diagram with a minimum of three moods. Draw circles for moods like Happy, Sad, Playful, Lethargic, or Depressed. Connect them with arrows to show how one mood can transition into another.

Label each arrow with the conditions that trigger the change. These conditions should now be based on multiple parameters.
Example:

- hunger > 80 AND boredom > 60 → Go to the "Depressed" mood.

This exercise will challenge you to think about how all of your parameters work together to create a single emotional state. It's the key to making your pet's behavior feel more realistic and complex.



Designing for Emotion: The Player's Journey



As a game designer, your goal is to create a strong bond between the player and their virtual pet. The EmoGBL framework is a simple, three-step process to help you do just that. It allows you to intentionally design for a specific feeling, instead of just hoping it happens.

Here is how you can use this framework to create an emotional connection:

- Desired Emotion: What do you want your player to feel? Decide on a core emotion you want to design for.
- Send Stimulus: These are the small moments in the game that will spark that emotion. It could be a specific animation of your pet looking sad, a unique sound it makes, or a simple pop-up message that says something like, "Your pet needs you."
- Answer & Reinforce: When a player responds to a stimulus (e.g., they feed a hungry pet), the game rewards them with a happy animation or a celebratory sound. This makes the player feel good, which encourages them to keep interacting with the pet and caring for it.

This process helps you build an emotional feedback loop that keeps players engaged and connected to their pet.

Reference: Plass et al., Emotional Game-Based Learning (2015).

Behavioral Matrix: Linking Moods & Activities

You've defined your pet's moods and activities. Now it's time to map out how those two things intersect. This table, called a behavioral matrix, is a critical tool for showing the player what your pet is feeling.

On the top, you have the moods you've defined. On the left, you have the activities your pet can perform. Your task is to fill in the cells with a specific action or reaction for each combination.

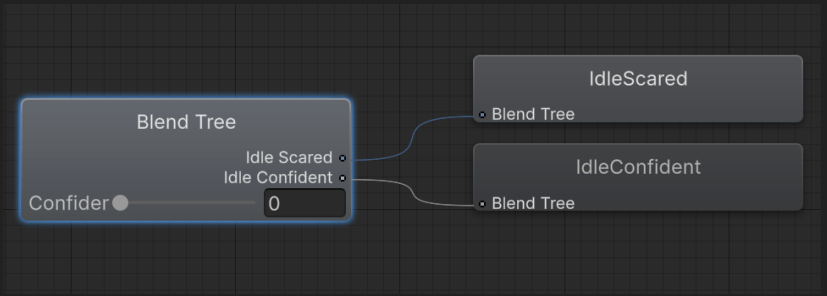
This exercise forces you to think about how a single action can change based on the pet's emotional state. This level of detail makes the pet's behavior more believable and less robotic.

Use the table below as a guide for your own pet.
The horizontal axis shows all the moods, and the vertical axis shows all the activities.



Controlling Moods with Blend Trees

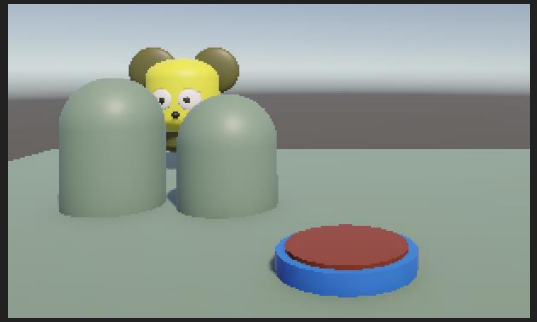
The Blend Tree is a powerful tool in Unity that allows you to smoothly transition between different animations based on a single parameter.



How does this apply to your pet?
Imagine your pet's Confidence is a value from 0 to 100.

- At a value of 0, the pet uses the Shy animations.
- At a value of 100, it uses the Confident animations.

As its confidence slowly increases from 0 to 100, the Blend Tree will gradually blend the animations. The pet won't just suddenly switch from hiding to playing; it will slowly and believably change its posture, movement, and expressions. This makes your pet's personality feel incredibly realistic and natural to the player.



How to setup a Blend Tree?

Step 1: Create the Blend Tree

- Open the Animator window for your character and right-click on an empty space, then select Create State > From New Blend Tree.
- Give the state a descriptive name, like "Idle".

Step 2: Set the "Parameter"

Click on the newly created Blend Tree.

In the Inspector window, under the Motion section, set the Blend Type to 1D.

Select the value that will control the blending from the parameter dropdown.

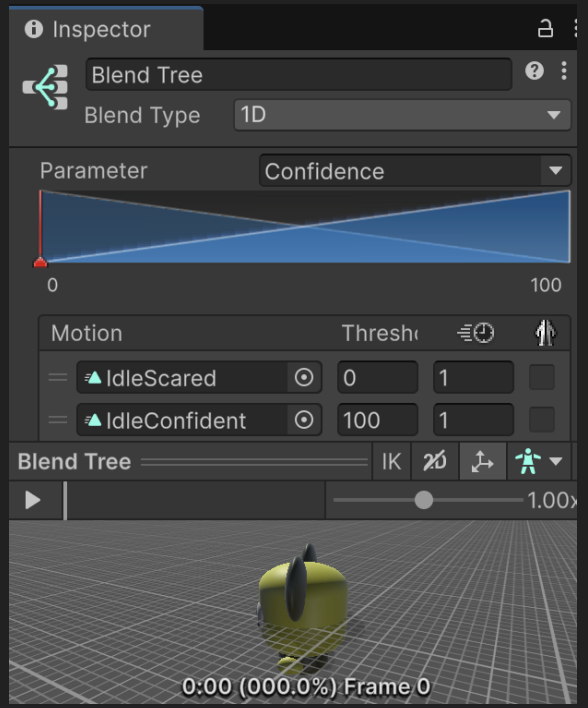
Step 3: Add two animation clips under "Motion"

- The Shy idle animation.
- The Confident idle animation.

Step 4: Check the Thresholds for these animations:

-Shy animation: 0

-Confident animation: 100



The Result

Now, when your pet's Confidence value is 0, it will use the Shy animation. When the value is 100, it will use the Confident animation. At any value in between (e.g., 50), Unity will automatically blend the two animations together, resulting in a smooth transition that reflects your pet's emotional state.

